Fiscal Year:	FY 2005	**Task Last Updated:**	FY 12/07/2004
PI Name:	Forgacs, Gabor Ph.D.		
Project Title:	Comparative Study Of Morphogenesis In 1g And Simulated Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture		
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City:	Columbia	**State:**	MO
Zip Code:	65211	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
Start Date:	01/01/2003	**End Date:**	12/31/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2005 per PI info in 12/2005 (previous end date was 12/31/2006)--1/2009		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG2-1611		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This interdisciplinary study is aimed at the better understanding of early developmental processes both in 1g and reduced gravity. Specifically, we investigate the morphogenetic phenomenon of cell sorting and follow and direct the self-assembly of organ modules of prescribed geometry. Sorting plays crucial role in the final physiological arrangement of tissues and organs. Cell sorting is one of the many self-assembly processes that take place in biological organisms. We are presently developing the technology to control and direct certain self-assembly processes. In particular, we are building tubular organ modules that eventually could be used as vascular or intestinal grafts. For this we use the evolving technology of organ printing, the computer controlled, three dimensional, layer-by-layer deposition of "bioink" particles, uniform-sized hetero- or homo-typical spherical cellular aggregates. The understanding and ability to reproducibly fabricate organ modules will be critical for long space flights and exporting life-form on Earth to other		

	planets. Understanding the effect of reduced gravity on bilogical self-assembly in general and on the early developmental process of cell sorting in particular. Understanding the effect of reduced gravity on early developmental processes will be critical for long space flights and exporting life-form on Earth to other planets.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The ability to engineer organ modules of specifc geometry will have significance for regenerative medicinal purposes, such as organ replacement.
Task Progress:	Up to date the most significant results of the project are 1. The explicit qualitative and quantitative demonstration that long exposure to reduced gravity has adverse consequences for the actin cytoskeleton of eukaryotic cells. 2. The modifications in the microfilament network due to reduced gravity, as far as morphology and tissue cohesivity are concerned can be reporduced by actin depolymerizing agents, such as Latrunculin A. 3. Morphogenetic, shape forming processes, such as cell sorting progress markedly differently in reduced gravity than in 1g. Such processes can be interpreted in terms of classical liquid phenomena. 4. Using the analogy between tissues composed of motile and adhesive cells and liquids, it is possible to fabricate "bioink", homo- or hetero-typical spherical cell aggregates, which can be delivered into scaffold gels by automatic delivery devices or bioprinters. This provides the possibility to develop the technology of "organ printing", a special branch of tissue engineering to build living structures of desired shape.
Bibliography Type:	Description: (Last Updated: 01/23/2009)
Articles in Peer-reviewed Journals	Jakab, K., Neagu, A., Mironov, V., Markwald, R.R. and Forgacs, G. "Engineering biological structures of prescribed shape using self-assembling multicellular systems" Proc. Natl. Acad. Sci. USA 101, 2864-2869 , Mar-2004
Articles in Peer-reviewed Journals	Forgacs, G., Hwang, I., SoonHyung, J., Janmey, P. and Burd, C. "The role of the cytoskeleton in signaling networks" J. Cell Sci. 117, 2769-2775 , Mar-2004
Articles in Peer-reviewed Journals	Newman, S. A., Forgacs, G., Hinner, B., Maier, C.W. and Sackmann, E. "Interaction of Fibers with Cell-sized Particles Alters Rheological Properties but not the Characteristics of the Gelation Transition in Model Connective Tissues" Physical Biology 1, 100-109 , Mar-2004
Articles in Peer-reviewed Journals	Jakab, K., Neagu, A. and Forgacs, G. "Organ printing: fiction or science" Biorheology 41, 371-375 , Jan-2004
Books/Book Chapters	Forgacs, G. and Foty, R.A. "Biological implications of tissue viscoelasticity" in "Function and regulation of cellular systems: experiments and models" Ed. A. Deutsch, M. Falcke, J. Howard and W. Zimmermann Birkhauser, Basel, 269-277 , Jan-2004